Please check the examination details below before entering your candidate information

Candidate surname	Other names

Group

Candidate Number

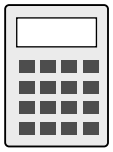
Elite IGCSE Academy | International GCSE

Expected Paper

November 2026

Time: **2 hours**

Paper reference **ELT/4MA1X**



Calculator allowed

Mathematics A

LINEAR · 4MA1

Higher Tier · Expected Exam

You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, group and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.

Information

- The total mark for this paper is 100.
- There are **26** questions in this paper.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Questions 19 to 26 are the most demanding. Aim to reach them with at least 45 minutes left. Aim to reach them with at least 45 minutes left.
- Check your answers if you have time at the end.

©2026 Elite IGCSE Academy

Written by Dr Eslam Ahmed. All rights reserved.

Dr Eslam Ahmed
Tel. **01120009622**
eliteigcse.com

Turn over ►

ELITE IGCSE ACADEMY

ELT 4MA1X 2026

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series

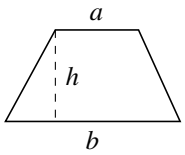
Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$

The quadratic equation

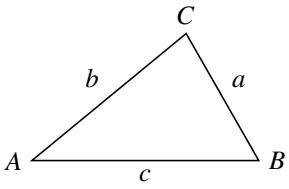
The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Area of trapezium $= \frac{1}{2}(a+b)h$



Trigonometry



In any triangle ABC

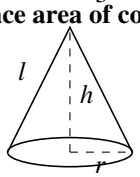
Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine Rule $a^2 = b^2 + c^2 - 2bc\cos A$

Area of triangle $= \frac{1}{2}ab\sin C$

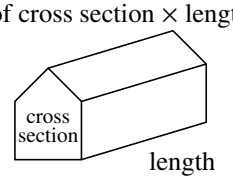
Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi r l$



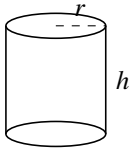
Volume of prism

= area of cross section × length



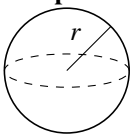
Volume of cylinder $= \pi r^2 h$

Curved surface area of cylinder $= 2\pi r h$



Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Without using a calculator, show that $2\frac{2}{3} \times 1\frac{1}{8} = 3$

Show each stage of your working.

(Total for Question 1 is 3 marks)

2 The price of a phone is increased by 12%.
The new price is then decreased by 12%.

Work out the overall percentage change in the price of the phone. State whether it is an increase or a decrease.

..... %

(3)

(Total for Question 2 is 3 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3

(a) Write 504 as a product of its prime factors. Show your working clearly.

. . . . . .

(2)

$180 = 2^2 \times 3^2 \times 5$

(b) Find the lowest common multiple (LCM) of 504 and 180

. . . . . .

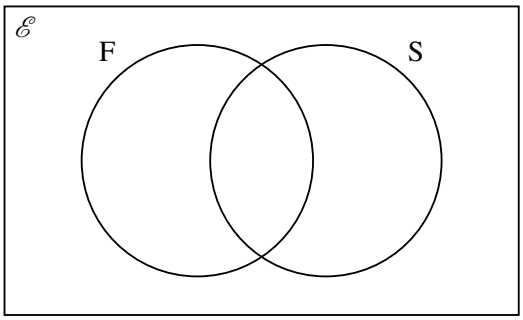
(1)

(Total for Question 3 is 3 marks)

---

4 There are 60 students in a year group.
35 students study French, 28 students study Spanish and 9 students study neither.

(a) Complete the Venn diagram to show this information.



(3)

(b) One of the students is chosen at random.
Find the probability that this student studies exactly one of the two languages.

. . . . . .

(1)

(Total for Question 4 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5 Solve the simultaneous equations

$$3x + 2y = 16 \qquad 5x - 3y = 14$$

Show clear algebraic working.

$x =$ .....

$y =$ .....

(3)

(Total for Question 5 is 3 marks)

---

6 Port B is 12 km from port A on a bearing of 065°.

(a) Work out how far east of A the port B is. Give your answer correct to 3 significant figures.

..... km

(2)

(b) Write down the bearing of A from B.

..... °

(1)

(Total for Question 6 is 3 marks)

---

7 Mina drives 120 km at an average speed of 80 km/h.
She then drives a further 90 km at an average speed of 60 km/h.
Work out her average speed for the whole journey.

..... km/h

(3)

(Total for Question 7 is 3 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

(a) Expand and simplify $(3x - 2)(4x + 5)$

.....

(2)

(b) Factorise fully $18x^2y - 12xy^3$

.....

(2)

(Total for Question 8 is 4 marks)

9

(a) Work out the value of $81^{\frac{3}{4}}$

.....

(1)

(b) Simplify fully $(27x^6)^{\frac{2}{3}}$

.....

(2)

(Total for Question 9 is 3 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

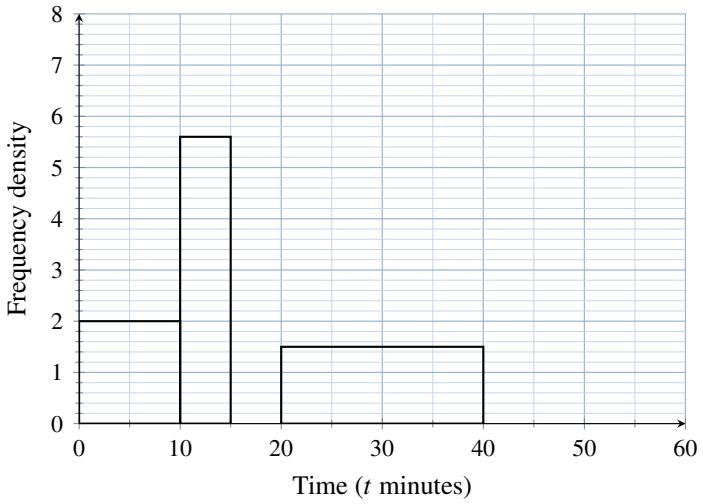
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Elite IGCSE Academy · Dr Eslam Ahmed · 01120009622 · eliteigcse.com

10 The table and the histogram give information about the times, t minutes, that some customers spent in a shop.

Time (t min)	Frequency
$0 < t \leqslant 10$	20
$10 < t \leqslant 15$	
$15 < t \leqslant 20$	36
$20 < t \leqslant 40$	
$40 < t \leqslant 60$	12



(a) Use the histogram to complete the table.

(2)

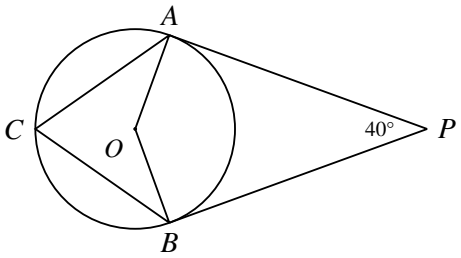
(b) Use the table to complete the histogram.

(2)

(Total for Question 10 is 4 marks)

11 A, B and C are points on a circle, centre O.
PA and PB are tangents to the circle.
Angle $APB = 40°$

Diagram **NOT** accurately drawn



(a) Work out the size of angle AOB. Give reasons for your answer.

.....°

(2)

(b) Work out the size of angle ACB. Give a reason for your answer.

.....°

(2)

(Total for Question 11 is 4 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

12 y is directly proportional to x^3
When $x = 2$, $y = 40$
Work out the value of y when $x = 3$

$y =$ .....

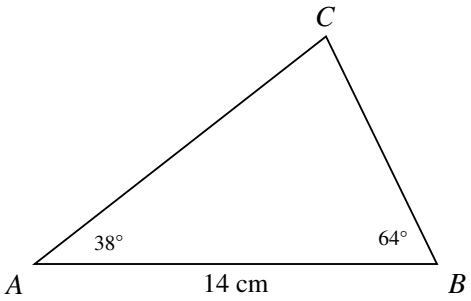
(3)

(Total for Question 12 is 3 marks)

---

13 In triangle ABC,
$AB = 14\,\text{cm}$, angle $BAC = 38°$ and angle $ABC = 64°$

Diagram **NOT** accurately drawn



(a) Work out the length of BC.
Give your answer correct to 3 significant figures.

..... cm

(3)

(b) Work out the area of triangle ABC. Give your answer correct to 3 significant figures.

..... cm^2

(2)

(Total for Question 13 is 5 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Elite IGCSE Academy · Dr Eslam Ahmed · 01120009622 · eliteigcse.com

14 A rectangle has length 8.4 cm, correct to 1 decimal place.
The width of the rectangle is 5 cm, correct to the nearest centimetre.

(a) Work out the upper bound for the area of the rectangle.

. . . . . cm^2

(2)

(b) Work out the lower bound for the length of a diagonal of the rectangle.
Give your answer correct to 3 significant figures.

. . . . . cm

(2)

(Total for Question 14 is 4 marks)

---

15 Express $\dfrac{3}{x+2} - \dfrac{x-4}{x^2+x-2}$ as a single fraction in its simplest form.

. . . . . .

(3)

(Total for Question 15 is 3 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

16

(a) Solve the inequality $2x^2 - 5x - 12 \leqslant 0$

.....

(3)

(b) How many integers satisfy the inequality in part (a)?

.....

(1)

(Total for Question 16 is 4 marks)

---

17 $f(x) = x^2 - 2$ $g(x) = 3x + 1$

(a) Find $gf(2)$

.....

(1)

(b) Solve the equation $fg(x) = 14$
Show clear algebraic working.

.....

(3)

(Total for Question 17 is 4 marks)

---

18 Use algebra to show that the recurring decimal $0.0\dot{7}\dot{2} = \frac{4}{55}$

(Total for Question 18 is 2 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

19 A particle moves along a straight line.
The displacement, s metres, of the particle from a fixed point O at time t seconds is

$$s = t^3 - 6t^2 + 9t$$

(a) Find the velocity of the particle when $t = 4$

. . . . . m/s
(2)

(b) Work out the total **distance** travelled by the particle between $t = 0$ and $t = 3$
Show your working clearly.

. . . . . m
(3)

(Total for Question 19 is 5 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

20 A bag contains n counters. 4 of the counters are white and the rest are black.
There are more black counters than white counters.

Aisha takes at random two counters from the bag without replacement.

The probability that the two counters are the same colour is $\frac{7}{15}$

Work out the value of n.
Show clear algebraic working.

$n =$ .....

(5)

(Total for Question 20 is 5 marks)

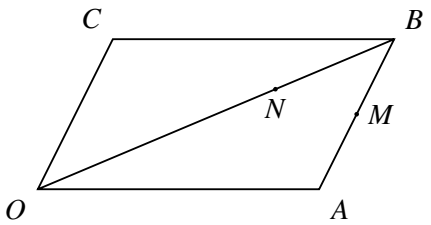
---

21 $OABC$ is a parallelogram.
$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$

M is the midpoint of AB.
N is the point on OB such that $ON : NB = 2 : 1$

Using a vector method, prove that C, N and M lie on a straight line, and find the ratio $CN : NM$.

Diagram **NOT** accurately drawn



$CN : NM =$ .....

(5)

(Total for Question 21 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

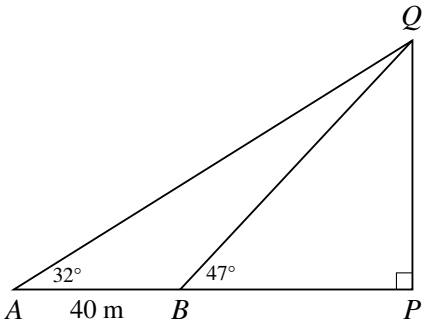
22 PQ is a vertical mast standing on horizontal ground at P.

A, B and P lie in a straight line on the ground, with $AB = 40$ m.

The angle of elevation of Q from A is 32°

The angle of elevation of Q from B is 47°

Diagram **NOT** accurately drawn



(a) Work out the height of the mast. Give your answer correct to 3 significant figures.

. . . . . m

(5)

(b) Work out the distance BP. Give your answer correct to 3 significant figures.

. . . . . m

(1)

(Total for Question 22 is 6 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

23

(a) Prove algebraically that the difference between the squares of any two consecutive odd numbers is a multiple of 8

(3)

(b) Give an example to show that this difference is not always a multiple of 16

(1)

(Total for Question 23 is 4 marks)

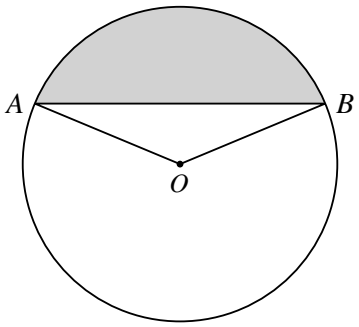
24

The diagram shows a circle, centre O, radius 13 cm.
AB is a chord of the circle and $AB = 24$ cm.

The shaded region is the minor segment between the chord AB and the arc AB.

Work out the area of the shaded region.
Give your answer correct to 3 significant figures.

Diagram **NOT** accurately drawn



. . . . . cm^2

(5)

(Total for Question 24 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

25 $f(x) = x^2 - 6x + 5$

(a) Write $f(x)$ in the form $(x - p)^2 - q$ where p and q are integers.

.....

(2)

The graph of $y = f(x + a) + b$ has its turning point at the origin.

(b) Find the value of a and the value of b.

$a =$ .....

$b =$ .....

(2)

(c) Write down the coordinates of the turning point of the graph of $y = f(2x)$

(..... , ...)

(1)

(Total for Question 25 is 5 marks)

---

26 Two containers, P and Q, are mathematically similar.
The surface area of P is $64\,\text{cm}^2$ and the surface area of Q is $100\,\text{cm}^2$.
Container Q holds 500 ml more than container P.
Work out how much container P holds. Give your answer correct to 3 significant figures.

..... ml

(5)

(Total for Question 26 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA